# Receiver cut mechanics and rookie NFL separation

*Research manuscript and technical appendix*

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## Abstract

I tested whether receiver movement measured during a Combine slot-strike drill improves prediction of rookie regular-season separation beyond conventional physical tests. The primary analysis used 59 receivers from the 2023–2025 draft cohorts and one analysis row per player. Nested ridge models held out entire draft years. Adding retention, early exit-speed ratio, entry speed and turn angle increased root mean squared error from 0.257 to 0.294 yards. A follow-up bootstrap that refitted and retuned both models over 300 player resamples produced an improvement interval spanning zero. Timing stress checks also exposed measurement sensitivity. These results support using movement profiles to form film-review questions, but do not establish a reliable pre-draft ranking tool.

This document combines the completed competition narrative with detailed measurement rules, complete model comparisons, a reproducibility record, and an assessment against the judging rubric. The primary analysis and later exploratory checks are identified separately throughout.

## Research record

Submitted Kaggle writeup
Public executable notebook

The study is an observational secondary analysis of competition data. The Word manuscript is editable and can be adapted to a course paper, poster or portfolio research item. Competition submission status does not imply peer review or publication.

# Part 1 Completed competition manuscript

## The scouting question

Two receivers can finish a drill in similar times and get there in different ways. One carries speed through the turn. Another slows down more, then has to build it back. I wanted to know whether those differences tell us anything useful about rookie NFL separation beyond the usual Combine tests.

**In my primary test, adding the movement profile did not consistently help.** Across 59 receivers held out by draft year, prediction error rose from **0.257 to 0.294 yards**. Some simpler profiles and smaller subsets improved, but those gains did not hold across the checks. I would use these measurements to organize film questions, not rank prospects.

## Why look inside the drill

A finish time hides where a player gained or lost speed. Research reports mixed evidence that conventional Combine measures predict later performance (Rishis et al., 2023). Change-of-direction work distinguishes turning demands from linear speed (Nimphius et al., 2016); deceleration research examines braking as its own demand (Harper et al., 2022). That makes the question worth testing. It does not make the answer positive.

## Who made it into the study

The data covers 2023–2025 prospects. I used NFL roster position WR: 106 receivers. The 107-player Combine WR testing group also includes one roster-listed tight end.

After an initial audit of 1,513 WR skill-drill sequences, I chose **SLOT_STRIKE_ROUTE_LEFT** before fitting NFL associations. Staying within one drill limits differences in what the movement asks a player to do. Of 101 sequences, 99 yielded eligible first-cut profiles from 99 players; two had no qualifying minimum.

NFL outcomes used draft-year regular-season routes with recorded separation at pass release, excluding nullified plays. There were 74 matched receivers before requiring at least 50 eligible observations. That left **59 players and 13,517 player-play records**: 24 from 2023, 19 from 2024, and 16 from 2025.

The effective sample is 59 receivers. Thousands of repeated plays do not turn it into thousands of independent prospects. The opportunity filter also selects players who got onto the field. In the broader rookie route cohort, separation was recorded for all 3,006 targets but only 11,727 of 13,838 non-target routes. Missingness remains a concern.

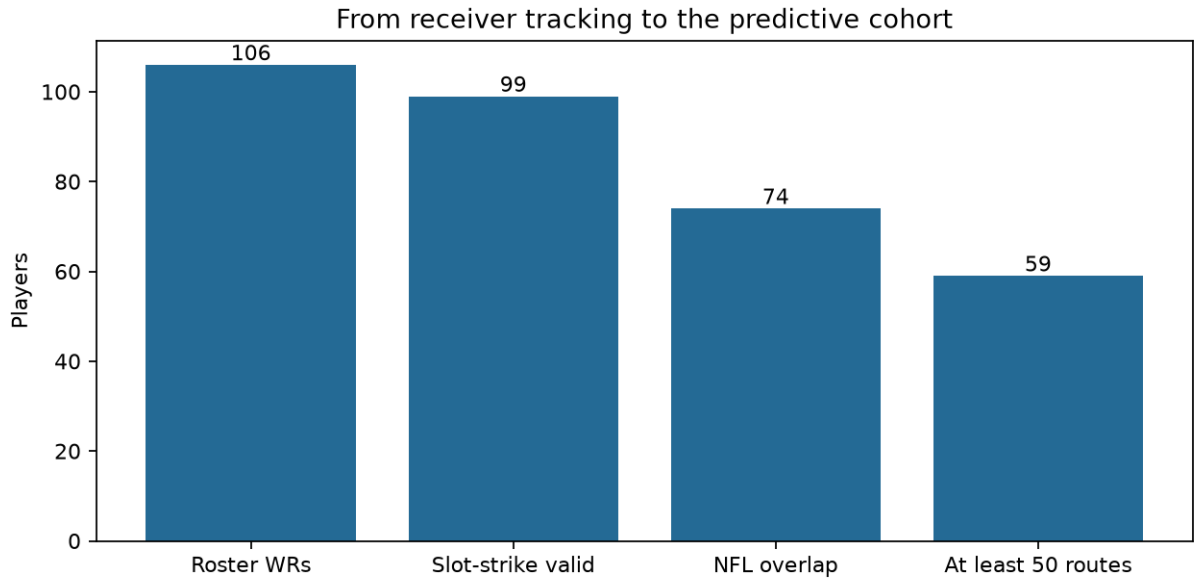



***Figure 1. How the available receiver pool becomes the predictive sample. Counts describe these competition records.***

## What happens at the cut

The detector smooths 10 Hz speed with a three-frame mean. It finds the first local minimum after an approach of at least 3 yards/second, a speed drop of at least 10%, and a 45–170 degree heading change. Approach speed is the maximum smoothed speed 1.2–0.2 seconds before the minimum. Heading windows are 0.8–0.3 seconds before and 0.3–0.8 seconds after. Coordinate-derived direction provides another geometry check.

My original idea was time to regain 80% of approach speed. The audit exposed a problem: **283 of 810 candidates (34.9%) already retained that much speed at the minimum**. Calling those zero-second recoveries would mix two different things.

The primary profile therefore uses speed retained at the minimum, speed at +0.5 seconds, approach speed, and turn angle. Both speed measures are divided by approach speed. Entry speed and angle describe the demand; this is not a pure reacceleration score.

Follow-up ends at the next detected braking turn, recording end, or two seconds. An interrupted +0.5-second endpoint is unavailable. Conditional 80% recovery stays descriptive. These are heuristic speed minima, not confirmed anatomical foot plants.

## A scouting example two different ways through the turn

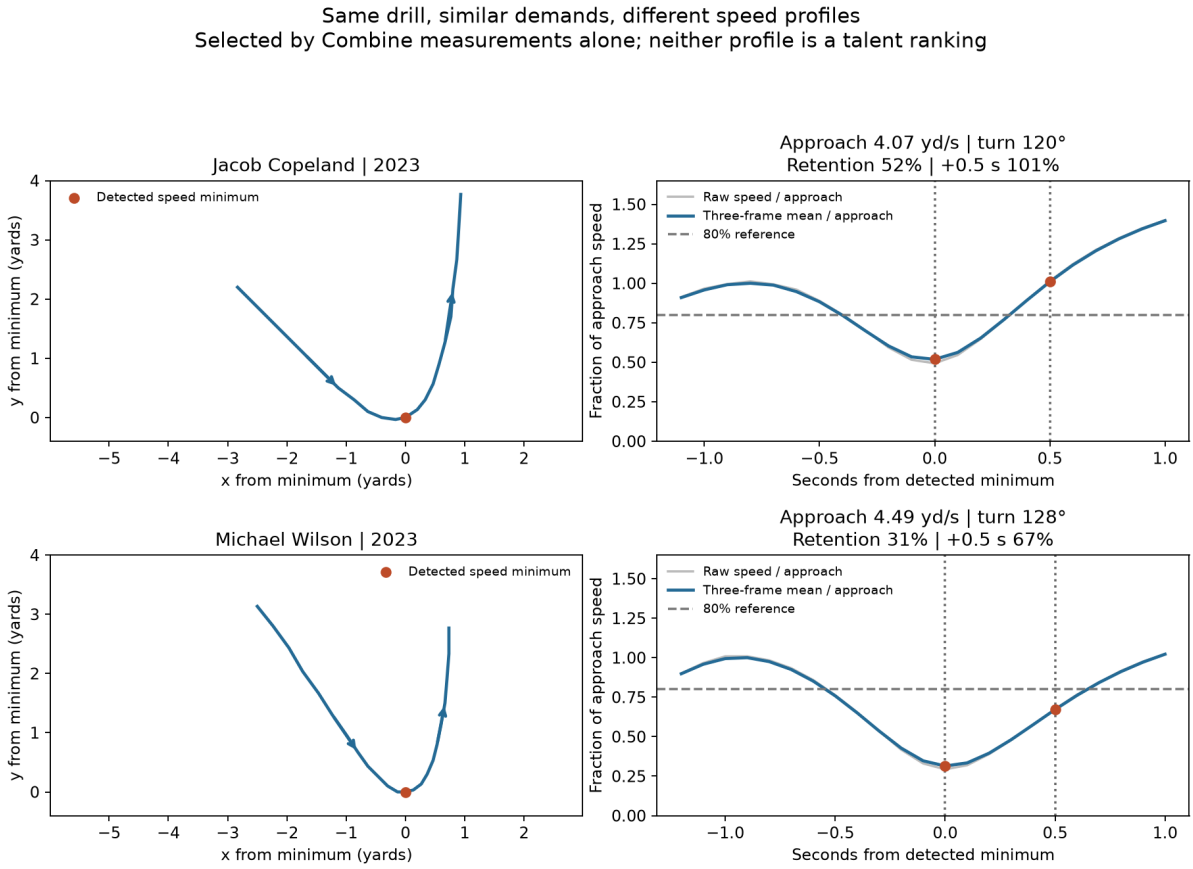



***Figure 2. Jacob Copeland and Michael Wilson in the same drill and 2023 cohort. Paths are centered on the detected minimum. Orange points mark the minimum and +0.5-second speed. Raw and smoothed curves are shown together.***

I selected this pair using Combine measurements alone: the largest retention contrast among same-year pairs within 0.5 yards/second of approach speed and 15 degrees of turn angle. NFL outcomes and playing opportunity were not used. This deliberately contrasting example illustrates the measurements; it is not a representative comparison or a controlled experiment.

Copeland approached at **4.07 yd/s**, turned **120 degrees**, retained **52%**, and reached **101%** of his approach reference at +0.5 seconds. Wilson approached at **4.49 yd/s**, turned **128 degrees**, retained **31%**, and reached **67%**. Their demands are similar, not identical. A lower entry speed can make retention easier; 101% simply means exceeding the earlier reference.

In a scouting meeting, I would use the figure to ask:

- Does the shallower slowdown recur on comparable game routes?
- Does deeper braking reflect the route instruction, a sharper stop, or defender leverage?
- Does either pattern actually create space against comparable coverage?

The next step is checking those questions on film. This drill does not contain NFL defender context, and one attempt cannot establish either player's ability or a training prescription.

## Does the profile improve predictions

The target is each receiver's mean observed separation at pass release. Ridge regression compares a baseline with a model adding the four movement features. The primary baseline includes height, weight, forty, ten-yard split, vertical and broad jumps, plus actual NFL route mix, coverage mix, and mean cushion.

**Actual NFL usage is future information at draft time.** The primary comparison asks whether tracking adds information once that context is known. The physical-tests-only comparison is the closer check of pre-draft relevance, and it also worsened with tracking.

Each outer fold held out an entire draft year. Four-fold validation within training players selected the ridge penalty from 0.1, 1, 10, and 100. Imputation and scaling were fitted within training folds. Players never crossed training/test partitions, and each receiver counted equally.

| Comparison | Players | Baseline error | With tracking |
|---|---|---|---|
| **Primary: physical tests + actual NFL context** | **59** | **0.257** | **0.294** |
| **Physical tests only** | **59** | **0.420** | **0.441** |
| Context + draft investment | 59 | 0.253 | 0.292 |
| At least 20 eligible routes | 65 | 0.251 | 0.289 |
| At least 100 eligible routes | 49 | 0.264 | 0.231 |
| Train 2023–2024, test 2025 | 16 | 0.218 | 0.203 |
| Retention and +0.5-second ratios only | 59 | 0.257 | 0.249 |

***Error is RMSE in yards of player-average separation; lower is better. It gives more weight to larger prediction misses. These are errors in average separation, not yards gained on a play.***

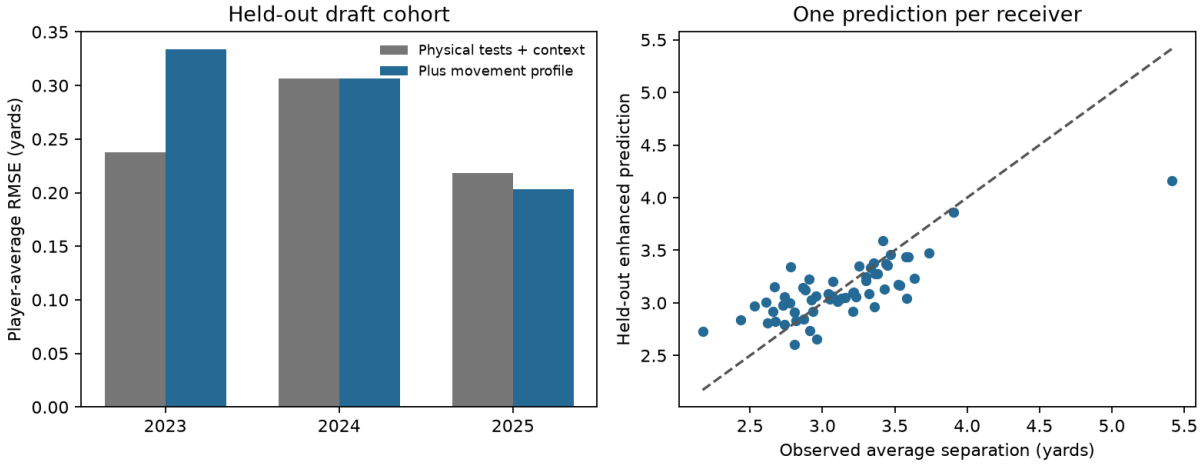



***Figure 3. Tracking worsened the 2023 holdout, barely changed 2024, and improved 2025. The chronological 2025 test is that same holdout, not independent replication.***

The primary error increase is **14.4%**. The original paired-player bootstrap, keeping predictions fixed, gave a baseline-minus-tracking difference of **−0.037 yards**, with a conditional 95% interval of **[−0.129, +0.031]**. Zero lies inside it.

I kept all prespecified checks, including alternative speed references and endpoints, in the notebook. Three-cone was missing for 40 of 59 players, leaving too few complete cases for the planned fit. I did not replace the primary result with the better-looking subset.

## Can I trust the measurements and the result

The original diagnostic review covered five trajectories per cohort before fitting outcomes. The follow-up adds **30 different profiles**, ten per cohort, selected without NFL values but reviewed after results were known. All showed plausible reversals and usable early recovery; no obvious early second maneuver required an exclusion.

This supports plausibility, not independent anatomical validation. Coordinate/report direction agreement is also an internal check of the same tracking system.

The timing stress test shifts the detected minimum by one or two 0.1-second frames and remeasures the ratios. One-frame shifts change retention by about **0.04** and the +0.5-second ratio by **0.11–0.12** in median absolute terms. Two-frame shifts reduce retention rank correlations to **0.74–0.75**. Clean-looking curves do not remove timing sensitivity. These deliberate shifts do not estimate the detector's actual error.

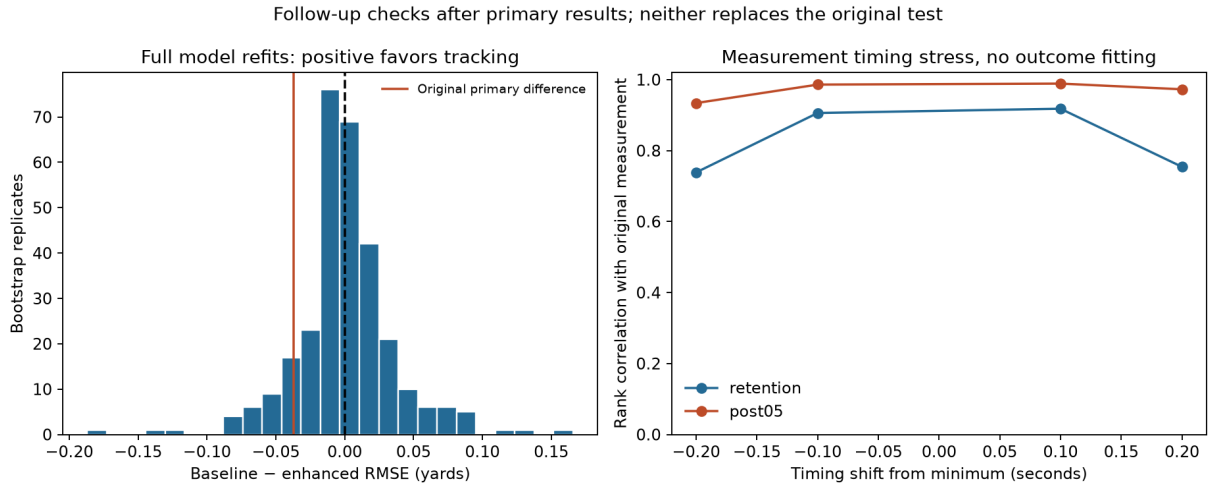



***Figure 4. Follow-up model refits and measurement timing stress. Positive error differences favor tracking. These checks were designed after the primary results were known.***

A second bootstrap resampled players within year **300 times**, refitting preprocessing and retuning both models each time. Duplicate copies of one player stayed together in inner folds. Its exploratory 95% percentile interval was **[−0.069, +0.083] yards**; tracking improved in **48%** of replicates. That fraction is a stability description, not a probability that tracking works. This check still conditions on three observed cohorts and fixed player measurements; it does not resample NFL plays or solve selection bias.

There were no repeated eligible primary-drill attempts per player, so test-retest reliability remains unknown. Single-attempt noise, missing outcomes, and unmeasured quarterback/team context still limit the study.

## What I would take into the film room

The useful distinction is between carrying speed through a turn and rebuilding it afterward. I would attach those profiles to comparable routes, record the film questions, and look for repeatable patterns before changing a grade.

The predictive claim stops here: **this profile did not consistently add information under the primary held-out test**. It does not show that movement never matters. Better repeated measurements and independent cohorts are needed before using it as a scouting score.

## Reproduce the study

The public Kaggle notebook regenerates the measurements, cohorts, predictions, all planned sensitivities, follow-up checks, and figures from the official competition files. It records package versions and input fingerprints. No GPU, external data, or credentials are needed, and raw competition tracking is not republished.

# Part 2 Technical research appendix

## Analysis design and data provenance

The primary protocol was specified before fitting Combine-to-NFL associations. The later scout example, expanded diagnostic review, timing perturbations and model-refit bootstrap were specified after the original results were known. Those checks supplement the original analysis; they do not constitute a new independent confirmatory test.

Five official inputs are used: players.csv, combine_results.csv, combine_tracking.csv, games.csv and player_play.csv. Player and game joins are checked for uniqueness. The outcome cohort requires roster WR, season equal to draft year, regular season, a recorded route, a non-nullified play and nonmissing separation at pass release. Outcome observations are aggregated to a player arithmetic mean. The primary opportunity minimum is 50 eligible routes.

## Measurement definitions

Primary question: does a tracking movement profile add held-out predictive information for rookie regular-season average separation at pass release, beyond conventional physical tests and realized NFL usage context?

Use one drill, SLOT_STRIKE_ROUTE_LEFT, to reduce cross-drill confounding. Use roster WRs, PLAYER sequences, regular finite 10 Hz records, and first qualifying directional speed minimum per attempt. Candidate rules retain the feasibility detector: three-frame mean, entry maximum in [-1.2,-0.2] seconds >=3 yd/s, >=10% speed loss, 45–170 degree circular heading change. Require sufficient before/after windows and coordinate-derived direction agreement. Flag the first subsequent sustained speed decline (three intervals, cumulative >=0.25 yd/s and heading change >=15 degrees); exclude the fixed 0.5s endpoint if that phase starts at/before it. Censor conditional T80 at this phase, end of recording, or two seconds. Do not treat already-above80 cases as instantaneous recovery.

Primary features: player median retention, speed ratio at +0.5s, entry speed, and turn angle across eligible first cuts. The latter two describe maneuver demand; the profile is not interpreted as pure reacceleration ability. Robust-entry median and +0.3s/+0.7s endpoints are measurement sensitivity checks. Plot a deterministic cohort-stratified sample blind to NFL outcomes. Report repeated-attempt coverage and repeatability limitations. No manual deletion based on NFL outcomes.

Primary outcome cohort: rookie season == draft year, REG only, recorded route/separation, non-nullified plays, >=50 eligible routes. One row per player. Target is that player's arithmetic mean observed separation. Baseline includes height, weight, forty, ten-yard split, vertical and broad jump plus route proportions, man/zone coverage proportions, and mean pre-snap cushion. The enhanced model adds the four movement-profile features. Realized route/coverage usage is post-draft contextual information: this primary comparison is conditional retrospective prediction, not a deployment-ready pre-draft forecast. Report physical-only and physical-plus-tracking comparisons separately.

## Model fitting and uncertainty

The primary baseline contains six physical measures and fixed route and coverage vocabulary proportions plus mean cushion. Missing values are imputed using training-set medians with missingness indicators; inputs are standardized within each training fold. The ridge penalty is selected from 0.1, 1, 10 and 100 by four-fold inner validation. The outer validation holds out one entire draft year at a time. Each receiver has one held-out prediction and receives equal metric weight.

The original interval uses 2,000 paired resamples of held-out players stratified by year, conditional on fixed predictions. The later 300-replicate bootstrap resamples the player analysis table within year and refits preprocessing and model tuning for each outer holdout. All copies of the same resampled player remain together

in inner folds. Its interval is an exploratory percentile stability summary, not a demonstrated population coverage guarantee. Neither method resolves uncertainty across unobserved cohorts, outcome missingness, measurement reliability or NFL opportunity selection.

## Complete model comparisons

| Analysis | N | Baseline RMSE | Tracking RMSE | Difference |
|---|---|---|---|---|
| Primary context | 59 | 0.2574 | 0.2945 | -0.0370 |
| Physical only | 59 | 0.4196 | 0.4412 | -0.0215 |
| Draft investment | 59 | 0.2529 | 0.2920 | -0.0391 |
| At least 20 routes | 65 | 0.2513 | 0.2891 | -0.0378 |
| At least 100 routes | 49 | 0.2643 | 0.2312 | +0.0331 |
| 2025 chronological | 16 | 0.2184 | 0.2030 | +0.0154 |
| Two ratios only | 59 | 0.2574 | 0.2493 | +0.0081 |
| Median approach speed | 59 | 0.2574 | 0.2929 | -0.0355 |
| Endpoint at 0.3 s | 59 | 0.2574 | 0.2963 | -0.0388 |
| Endpoint at 0.7 s | 59 | 0.2574 | 0.2482 | +0.0092 |

***RMSE and differences are in yards. Difference is baseline minus tracking; positive favors tracking. All planned comparisons remain visible regardless of direction.***

## Timing stress results

| Shift seconds | Feature | Players | Median absolute change | Rank correlation |
|---|---|---|---|---|
| -0.2 | retention | 99 | 0.145 | 0.739 |
| -0.2 | post05 | 99 | 0.252 | 0.934 |
| -0.1 | retention | 99 | 0.038 | 0.906 |
| -0.1 | post05 | 99 | 0.121 | 0.986 |
| +0.1 | retention | 99 | 0.043 | 0.918 |
| +0.1 | post05 | 99 | 0.107 | 0.989 |
| +0.2 | retention | 99 | 0.156 | 0.754 |
| +0.2 | post05 | 99 | 0.200 | 0.973 |

***Both feature measures are dimensionless ratios. Each timing shift is measured against its original value; these are stress tests rather than estimates of actual detector error.***

## Reproducibility record

The self-contained notebook embeds both analysis implementations and protocols, regenerates all outputs, and records input SHA256 fingerprints. Local execution passed every code cell. The latest saved public Kaggle notebook was Version 3 with script version 363064012, completing successfully in 162.5 seconds. The earlier Version 2 completed in 230.1 seconds. Different runtimes do not imply different sample sizes or primary conclusions.

Local package record: python 3.12.14, pandas 2.2.3, numpy 2.3.5, sklearn 1.9.1, matplotlib 3.11.2. Kaggle used Python 3.13.15, pandas 2.3.3, NumPy 2.1.3, scikit-learn 1.6.1 and Matplotlib 3.10.0 in the verified Version 2 run.

Audit checks cover player uniqueness, fold coverage, metric reconciliation, measurement accounting, bootstrap replicate counts and scout-pair matching. Scripts are scripts/predict_receiver_separation.py, scripts/strengthen_study.py and scripts/build_notebook.py in the project workspace. These local paths document provenance; the public notebook is the portable executable research artifact. Raw competition tracking is read in place and is not bundled in this document.

## Research contributions and limits

The contribution is a transparent measurement and validation workflow linking one Combine drill to rookie NFL separation, with a concrete demonstration of the retention-versus-recovery distinction. It is not a claim to invent change-of-direction analysis. The central result is a limit on this feature set and sample, not proof that movement mechanics have no value.

The primary model uses actual NFL context and cannot be deployed unchanged before a draft. The 59-player sample is selected by opportunity, contains only three cohorts, and has no eligible within-player repeat attempts for reliability estimation. Average separation at pass release captures only one part of receiver performance. Team, quarterback, route instruction, defender behavior and missing outcomes remain incompletely addressed. Later checks are post-result diagnostics and cannot erase those limitations.

# Part 3 Judging assessment and research priorities

This is an editorial assessment against the published rubric, not a prediction of the judges' scores or a ranking against other entries. The improved submission is stronger in communication, transparency and diagnostic depth. Its main competitive gap is still a validated football application with repeatable value.

| Criterion | Weight | Current position |
|---|---|---|
| Football | 30% | Clear application question and scout example; practical benefit remains unvalidated |
| Data science | 30% | Reproducible and candid; small selected sample and timing sensitivity limit inference |
| Writing | 20% | Direct first-person narrative with clear results and limits |
| Visualization | 20% | Four informative inline figures; polished diagnostics but no validated operational tool |

## Biggest remaining weaknesses

Football usefulness is the largest gap. The case study generates sensible film questions, but it does not demonstrate that a scout makes a better decision or that movement profiles translate reliably to game cuts. A stronger next study would define comparable game-route situations and an outcome-independent annotation procedure before viewing associations.

Measurement reliability is unresolved. One attempt per player cannot separate stable traits from execution noise. The timing stress check shows why this matters: a one-frame change materially shifts speed ratios. Independent annotation by a second reviewer and repeated eligible measurements would help; neither can be claimed from the current diagnostic plots.

Generalization is uncertain. Three cohorts and 59 selected players leave the estimates fragile. The physical-only comparison worsens, while favorable sensitivities change with sample or definition. A newly locked analysis on genuinely new data would be stronger evidence than further tuning on this dataset.

Distinctiveness is moderate. Receiver change of direction and separation are explicitly aligned with the competition theme. The retention/recovery distinction and honest validation are useful, but a winning entry may need a more distinctive football insight or a demonstrated evaluation workflow. More complex modeling alone would not fill that gap.

## Priorities for the next research version

1. Define a prospective film-review study with matched route demands, a fixed annotation rubric and a measurable decision or performance endpoint.
2. Obtain independent cut annotations and repeated tracking attempts where legitimately available; quantify agreement and repeatability before ranking players.
3. Lock the metric and model before testing on a genuinely new cohort. Report negative results and all exclusions.
4. If only current data are available, improve explanation and error analysis rather than promoting the best-performing sensitivity.

## Suggested research portfolio description

Developed a reproducible study linking NFL Combine receiver tracking to rookie regular-season separation. Built drill-specific cut measurements, evaluated nested cohort-held-out ridge models on 59 receivers, and investigated timing sensitivity and model stability with 300 refits. Found no consistent added predictive value and translated the measurement results into targeted scouting film questions.

# References and research links

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